IBM **CICS**
Transaction Server for z/OS

CICS Sample GLUE : XFCREQ

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Content

The following are described in this presentation:

- An overview of CICS exits
- The problem we are seeking to resolve
- The solution
- The artifacts
- Solution alternatives

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Overview

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overview

Those familiar with the systems management of CICS TS will be aware of the ability to execute user programs at strategic points during the execution of a transaction. The intent being that user code can modify the behaviour of CICS to meet specific requirements.

CICS provides different exit types, which serve different purposes (see Customizing with user exit programs - IBM Documentation).

Global User Exits (GLUEs) are perhaps the most well known (see Global user exit programs - IBM Documentation).

This document, the scenario it describes, and the business problem it seeks to resolve use the GLUE interface at the XFCREQ exit point.

The XFCREQ exit points is driven for file control API requests before any action is taken by CICS file control; this allows the exit to modify the input parameter list, including the file name being accessed.

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overview

The package of programs, CICS resource and VSAM file definitions provided are sufficient to demonstrate the solution.

We have also described several modifications to the design which may better suit your scenario and will hopefully provide some additional flexibility in implementation.

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Terminology

Certain terms are used in this document which are described below. Clarifying their meaning will hopefully reduce ambiguity.

- Application – A set of CICS resources which represent a business application. Typically made up of CICS transaction, program and data resource definitions.
- TRANSACTION – A 1-4-character name of a CICS TRANSACTION definition used to initiate a business function in CICS.
- PROGRAM – A 1-8-character name of a CICS PROGRAM definition. Executed as part of a transaction.
- Task – a single executing instance of a transaction.
- FILENAME – A 1-8-character name of a CICS FILE resource definition. Contains the name of the underlying VSAM dataset to be accessed.
- DATASET – The underlying dataset name defined to z/OS. Up to 44 character in length.

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The problem to be solved

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The problem

Business applications almost always require access to data. CICS provides access to many different data sources, of which one of the most frequently used are VSAM datasets.

Without going into detail as to why VSAM datasets are used or the enhancements to VSAM which go a long way to addressing the problem described, many applications still have a requirement to periodically suspend CICS access to VSAM datasets in order that housekeeping activities can occur.

Housekeeping could be reorganisation of the files or the updating of data with information gathered outside of CICS.

This suspension of service is known as an outage and in some cases can be a daily occurrence. Clearly, for highly available applications, this has a negative impact and the ability to reduce or remove the outage is advantageous.

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The scenario

- A finance application is accessed in CICS by invoking transaction ABCD.
- Transaction ABCD may access one or more of the following four FILEs named:
  - SMPFILE1, SMPFILE2, SMPFILE3 and SMPFILE4.
- The four FILE definitions map to four datasets as follows:
  - SMPFILE1 - FINANCE.DATASET1
  - SMPFILE2 - FINANCE.DATASET2
  - SMPFILE3 - FINANCE.DATASET3
  - SMPFILE4 - FINANCE.DATASET4

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ABCD task starts

- Executes initial business logic
- EXEC CICS READ FILE(SMPFILE1)
- Analyse data received
- EXEC CICS READ FILE(SMPFILE2)
- EXEC CICS READ FILE(SMFFILE4)
- Execute business logic
- EXEC CICS READ UPDATE FILE(SMPFILE3)
- Execute business logic
- EXEC CICS REWRITE FILE(SMPFILE3)
- EXEC CICS WRITE FILE(SMPFILE4)

ABCD task ends

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ABCD task starts

- Executes initial business logic
- EXEC CICS READ FILE(SMPFILE1) - CICS reads from FINANCE.DATASET1
- Analyse data received
- EXEC CICS READ FILE(SMPFILE2) - CICS reads from FINANCE.DATASET2
- EXEC CICS READ FILE(SMFFILE4) - CICS reads from FINANCE.DATASET4
- Execute business logic
- EXEC CICS READ UPDATE FILE(SMPFILE3) - CICS reads(update) from FINANCE.DATASET3
- Execute business logic
- EXEC CICS REWRITE FILE(SMPFILE3) - CICS writes to FINANCE.DATASET3
- EXEC CICS WRITE FILE(SMPFILE4) - CICS writes to FINANCE.DATASET4

ABCD task ends

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The scenario

Currently, on a daily basis, access to transaction ABCD is disabled in order that files SMPFILE1, SMPFILE2 and SMPFILE3 can be closed in CICS, allowing a new version of the underlying datasets to be created.

Note: SMPFILE4 contains static data which does require updating on a daily basis.

FINANCE.DATASET1-3 are deleted and recreated using updates from batch processing which have been running in parallel. The exact processes involved in creating the new instances will vary from customer to customer and are not covered here. Once the process is complete, and the new datasets are ready and available, the FILENAMEs can be opened in CICS and access to transaction ABCD can resume.

Changing the FILENAME used by the application is possible but isn't generally a feasible solution. This is because most companies resist changes to application code when alternatives are available.

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Sequence of events

A possible sequence of events that occur during outage

1. IN CICS: Application disabled
   - Business application no longer available to users
2. IN CICS: FILENAMEs SMPFILE1-3 CLOSED and DISABLED
3. IN BATCH: DATASETS FINANCE.DATASET1-3.NEW created (could be done prior to 1)
4. IN BATCH: DATASETS FINANCE.DATASET1-3 deleted
5. IN BATCH: DATASETS FINANCE.DATASET1-3.NEW renamed to FINANCE.DATASET1-3
6. IN CICS: FILENAMES SMPFILE1-3 OPENED and ENABLED
7. IN CICS: Application enabled
   - Business application available to users

Between points 1 to 7 an application outage is occurring.

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Requirements of the application

The finance application has some requirements that must be observed:

- When the application is disabled (1) currently running tasks must be allowed to complete normally
- The application runs across multiple CICS regions and tasks in all CICS regions must complete as part of step 2.
  - This has implications for VSAM RLS and non-RLS datasets, but the proposed solution works for both access types.
  - Step 2 would involve a QUIESCE activity for VSAM RLS datasets
  - Step 6 would involve an UNQUIESCE activity for VSAM RLS datasets
- No changes to the application code or the FILENAME definitions they specify when accessing data
- If multiple copies of datasets are to be used, then a single instance of a task must be consistent in which copy or set of copies they access.

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Proposed solution

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Overview of a solution

CICS FILE definitions map to a single underlying dataset as is the case for FILENAME SMPFILE1 which maps to z/OS dataset FINANCE.DATASET1.

This solution implements a mechanism whereby eligible files used by the finance application are intercepted and mapped to one of two underlying filenames. The application is unaware that this is happening thus allowing different datasets to be seamlessly removed or made available to the application.

To allow this process to occur for a group of files, the concept of a FILESET is introduced which defines a set of one or more files which must be consistent in their content. This allows a FILESET indicator to be used to determine which set of files should be used at any given time.

The XFCREQ GLUE point is used to implement this solution. Processing within the GLUE determines which FILESET to use when a task accesses its first file. The FILESET is stored for the life of the task thus ensuring a switch of FILESETs while a task is running does not cause more than one FILESET to be used.

It should be noted that multiple programs can be active at the XFCREQ exit point and that when activated the exit is invoked for **ALL** file control requests, irrespective of application or filename.

It should also be understood that the exit is invoked in the CICS region where the EXEC CICS file access occurs, not the CICS region where the FILE is located. This is relevant for infrastructures with AOR-FOR configurations.

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Processing <u>without</u> the exit active

ABCD task starts

- Executes initial business logic
- EXEC CICS READ FILE(SMPFILE1)

<Program suspends>

CICS FILE CONTROL invoke passing in parameters:
- FILENAME
- Key of record
- Input/output area
- Other parameters

Note: the actual file request may be routed to another CICS regions. Not before the XFCREQ GLUE has been invoked (if active in the regions).

Dataset request processed

RESPONSE RETURNED TO TASK:

<Program resumes>

- Processing continues

ABCD task ends

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Changes to CICS

Changes to CICS

CICS FILE definitions REMOVED:

- SMPFILE1 - FINANCE.DATASET1
- SMPFILE2 - FINANCE.DATASET2
- SMPFILE3 - FINANCE.DATASET3
- Note: SMPFILE4 remains as not part of daily housekeeping

Two sets of new FILES definitions CREATED:

- £MPFILE1 - FINANCE.DATASET1.V1 (FILESET A)
- £MPFILE2 - FINANCE.DATASET2.V1 (FILESET A)
- £MPFILE3 - FINANCE.DATASET3.V1 (FILESET A)

- #MPFILE1 - FINANCE.DATASET1.V2 (FILESET B)
- #MPFILE2 - FINANCE.DATASET2.V2 (FILESET B)
- #MPFILE3 - FINANCE.DATASET3.V2 (FILESET B)

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Point in time switching

The exit has been written to support point-in-time switching. IF enabled, the exit will use a primary FILESET for requests before the specified point in time and switch to the secondary FILESET when it determines the task started after the point in time specified in the Global work area.

A Flag in the GWA call GWASWITCH controls whether point in time switching is active.

- GWASWITCH = Y – Time based switching is active
  - Based on task start time, use GWAFSET prior to GWATIME and GWAFSETN) after GWATIME
- GWASWITCH = N – Time based switching is inactive. Use the FILESET specified by GWAFSET
  - The GWAFSET value is examined the first time the XFCREQ exit is driven.

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Processing with the exit active

ABCD task starts

- Executes initial business logic
- EXEC CICS READ FILE(SMPFILE1)

<Program suspends>

CICS FILE CONTROL invoke passing in parameters:
- FILENAME
- Key of record
- Input/output area
- Other parameters

XFCREQ exit called passing parameter list.
- Exit program checks if first file request by task
  - Checks GWA to determine FILESET to use (A)
  - FILESET A saved in task specific area
- Scan list of eligible files
  - Find SMPFILE1 in list.
  - Check FILESET to use (A)
  - Changes FILE name to £MPFILE1

<Program resumes>

- Processing continues

ABCD task ends

RESPONSE RETURNED TO TASK:
Note: Task is unaware that the request has been redirected to £MPFILE1.

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ABCD task starts

- Executes initial business logic
- EXEC CICS READ FILE(SMPFILE1) - Redirected to £MPFILE1
- Analyse data received
- EXEC CICS READ FILE(SMPFILE2) - Redirected to £MPFILE2
- EXEC CICS READ FILE(SMFFILE4) - not in list so not redirected
- Execute business logic
- EXEC CICS READ UPDATE FILE(SMPFILE3) - Redirected to £MPFILE3
- Execute business logic
- EXEC CICS REWRITE FILE(SMPFILE3) - Redirected to £MPFILE3
- EXEC CICS WRITE FILE(SMPFILE4) - not in list so not redirected
-

ABCD task ends

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notes

- Multiple programs can be enabled at the XFCREQ exit point, so coexistence is necessary.
  - The exit was tested with multiple programs enabled at the XFCREQ exit point
- Each program enabled at an exit point can request a Global Work Area (GWA)
- The GWA is defined for the total period the Exit program is enabled.
- The GWA must be initialised between the ENABLE and START of the program at the exit point
- For this design, the FILESET to use, the substitution character used to change the FILENAME , and the list of eligible files are all stored in the GWA.
- A maintenance program is supplied to ENABLE and DISABLE the GLUE and to DISPLAY and UPDATE the FILESET in use in the GWA.
- Once ENABLED and STARTED, the exit program is invoked for all file control requests in the region.

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How does this impact the housekeeping window

This was the before sequence of events

1. IN CICS: Application disabled
   - Business application no longer available to users
2. IN CICS: FILENAMEs SMPFILE1-3 CLOSED and DISABLED
3. IN BATCH: DATASETS FINANCE.DATASET1-3.NEW created (could be done prior to 1)
4. IN BATCH: DATASETS FINANCE.DATASET1-3 deleted
5. IN BATCH: DATASETS FINANCE.DATASET1-3.NEW renamed to FINANCE.DATASET1-3
6. IN CICS: FILENAMES SMPFILE1-3 OPENED and ENABLED
7. IN CICS: Application enabled
   - Business application available to users

From point 1 to 7 an application outage is occurring.

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How does this impact the housekeeping window

This is the after sequence of events

1. IN CICS: Applications using FILESET A (Directed to .V1 datasets by using £MPFILE definitions)
2. IN BATCH: DATASETS FINANCE.DATASET1-3.V2 created
3. IN CICS: FILENAMES #MPFILE1-3 OPENED AND ENABLED
4. IN CICS: UPDATE GWA to specify FILESET B (Direct by #MPFILE definitions to .V2 datasets)
   - New tasks will start using .V2 files
5. IN CICS: £MPFILE1-3 CLOSED and DISABLED
   - Tasks started prior to 4 allowed to complete normally
6. IN BATCH: DATASETS FINANCE.DATASET1-3.V1 now available for batch processing

Notes:

- No outage is taken by the CICS application during this period.
- The processing would be reversed on the following day- switch would be from .V2 to .V1

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Limitations of the solution

The following limitations exist with this solution:

- The creation of .V1 and .V2 datasets are not dependent on each other
  - This means the creation of .V2 does not require .V1 as input and vice versa
- A brief time exists where both sets of files are being used, albeit by different tasks
  - This must be tolerable by the application
- With time-based switching the switch over time can be coordinated across CICS regions and LPARS in the same SYSPLEX (assuming the CICS times are synchronised)

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Artifacts

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The following artifacts are provided

More details on each are provided in the following slides

- This documentation file
- Assembler program SMPFCREQ (source)
  - Assembler copybooks (GWAMAP, EXITDATA and UEPTSMAP)
- COBOL program EXITMAIN (source)
  - COBOL copybooks (GWAMAP, EXITDATA, UEPTSMAP, HEXTRAN, HEXTRANL)
- DFHCSDUP input for the following definitions (split into two files, Local definitions and remote definitions)
  - PROGRAM definitions for SMPFCREQ and EXITMAIN
  - TRANSACTION definitions MAIN (uses EXIMAIN)
  - FILE definitions for £MPFILE1-4 and #MPFILE1-4
- JCL to execute IDCAMS
  - Create the sample FINANCE.DATASETn.Vn
  - Load files with some sample records
- PLT definition for PLTPI enablement of the exit
  - Note EXITMAIN requires data to be specified in the INITPARM= SIT parameter
- SIT PARAMETER INITPARM

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SMPFCREQ

Description of the program and its control

This sample assembler program is written to be enabled at the XFCREQ EXIT POINT.

It checks a GWA is present of the expected length.

It's important that the assembler and COBOL version of the following DSECTS/COPYBOOKS are kept in sync:

- GWAMAP – Maps the Global Work Area
- EXITDATA – contains constants used by EXITMAIN and SMPFCREQ to initialise and set the GWA and UEPTSTOK structures
- UEPTSMAP – Maps the area of storage used by SMPFCREQ to store task specific information.
  - This structure is created the first time a task issues a file control request
  - The structure is key to ensuring the task uses the same FILESET for the entire period it executes.

SMPFCREQ contains numerous WTO messages to aid testing and debugging. It also contains a subroutine to display data in hexadecimal format. Most of these can be removed once alterations, testing and debugging are complete.

Details on assembling and linking of GLUEs is described in the IBM DOC.

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GWAPMAP

Assembler and COBOL

MAPS the GWA created as part of the EXEC CICS ENABLE PROGRAM command.

Fields are:

- GWAEYE – Eyecatcher to aid debugging
- GWAFSET – Indictor of which FILESET to use. Currently will contain ‘A’ or ‘B’
  - Note, if GWASWITCH is Y, GWAFSET is used prior to the reference time.
- GWSWITCH – Y|N – whether time-based switching is active.
- GWAFSETN – only used if GWASWITCH=Y. Is the FILESET to use for tasks starting after the GWATIME
- GWATIME – 13-byte time stamp in format HHMMSS.TTTTTT.
- GWAFLIST – an array of 8-byte FILE names which are eligible for switching. It is recommended that the most heavily accessed files are first in the list. If a file name isn’t found in the list, the filename specified in the API command is passed unaltered to CICS file control.
- GWAFLLEN – Length of the file array
- GWALEN – length of the GWA structure
- GWAFiLEN – the number of files in the file array.

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EXITDATA

Assembler and COBOL

This DESECT/COPYBOOK contains data constants used by the SMPFCREQ and EXITMAIN programs to populate the GWA and task user token structures.

Fields are:

- UEPPGMNM – Program name of the sample program, checked when searching for the task token structure
- UEPFSETA – FILESET A indicator
- UEPFSETB – FILESET B indicator
- UEPEYE – task token structure eyecatcher
- GWAEYE – GWA structure eyecatcher
- GWAFSETA – GWA FILESET A indicator
- GWAFSETB – GWA FILESET B indicator
- FILELIST – List of eligible files which is copied into the GWA as part of exit initialisation.
  - It is recommended that the most heavily accessed eligible files appear at the top of the list. This will reduce to search time to locate the file.
- FILELLEN – Length of the file list

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UEPTSMAP

Assembler and COBOL

This DESECT/COPYBOOK maps the task token structure referenced by the UEPTSTOK field in the exit parameter list.

On first entry UEPTSTOK will contain the address of EISEXITT which will contain 0. When this task token area is created, its address will be stored in EISEXITT or the chain reference by it.

See Using the task token UEPTSTOK - IBM Documentation for details on the use of this field, including use by multiple programs at the same exit point.

Fields are:

- UEPNEXT – address of the next exit block in the chain
- UEPPROG – name of the program owning this structure (should match the SMPFCREQ program name)
- UEPEYE – Exit block eyecatcher
- UEPFSET – FILESET in use by this task
- UEPFCHAR – character used to replace first character of eligible file (based on FILESET)
- UEPSTIME – Time this task was started (Taken from INQUIRE ASSOCIATION STARTTIME request)
- UEPLFILE – Last filename specified by task on file request
- UEPNFILE – new file name after alteration based on FILESET substitution

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EXITMAIN

Program to perform exit enable, disable, display and update

This sample program can be invoked via a console, 3270 display or directly via the PLT.

If invoked by PLT, an INITPARM SIT parameter must be specified for the EXITMAIN program.

The program accepts commands:

- ENABLE <FILESET> <TIME BASED> <ALTERNATE FILESET> <REFERENCE TIME>
- DISABLE
- DISPLAY
- UPDATE <FILESET> <TIME BASED> <ALTERNATE FILESET> <REFERENCE TIME>

- FILESET is a single character ‘A’ or ‘B’
- TIME BASED is a single character ‘Y’ or ‘N’
- ALTERNATE FILESET is a single character ‘A’ or ‘B’
- REFERENCE TIME is a 13-character string in format HHMMSS.TTTTTT

The exit will issue status messages and write to a TS queue call EADEDIAG (if DEBUG is specified).

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EXITMAIN

Invoking the program

Invoking as part of PLTPI processing

- ENABLE A Y B 120000.000000 – Will enable the exit and initialise the GWA to use FILESET A for tasks starting before midday, after midday FILESET B will be used
- ENABLE B N B 235959.999999 - Will enable the exit and initialise the GWA to use FILESET B

Invoking via console command

F <CICSNAME>,MAIN ENABLE - See format for PLTPI invocation

F <CICSNAME>, MAIN ENABLE - See format for PLTPI invocation

F <CICSNAME>, MAIN DISABLE - Will STOP and DISABLE to EXIT

F <CICSNAME>, MAIN DISPLAY - Will DISPLAY the current FILESETs and reference time in use by the exit

F <CICSNAME>, MAIN UPDATE - Same command syntax as ENABLE

F <CICSNAME>, MAIN UPDATE - Same command syntax as ENABLE

Invoking via 3270

MAIN <parameters> - the same parameters as specified via the console

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CSD LOCAL

DFHCSDUP input for the FOR regions

Contains local file definitions for the six files used in this example.

The DSNAME values should be changed to name which conform to your site standards.

If the exit and file in the same CICS region, these definitions should be installed.

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CSD-REMOTE

DFHCSDUP input for the local AOR CICS Region where the EXIT will run

Contains the program and transaction definitions for the exit and maintenance programs.

This file also contains remote definitions used in an AOR-FOR configuration. The REMOTESYSTEM name of the FOR will need to be changed to match your configuration.

If the exit and files are to reside in the same CICS Region, the FILE definitions in this file should not be used.

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FILES

DFHCSDUP input for the local AOR CICS Region where the EXIT will run

Contains the program and transaction definitions for the exit and maintenance programs.

This file also contains remote definitions used in an AOR-FOR configuration. The REMOTESYSTEM name of the FOR will need to be changed to match your configuration.

If the exit and files are to reside in the same CICS Region, the FILE definitions in this file should not be used.

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Solution alternatives

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Some potential changes to meet alternative requirements

More than two FILESET solution

The proposed solution uses substitution of the first character of the FILENAME to derive the new FILENAME

- Two possible values are available determined by the GWAFSET, with UEPFCHAR providing the substitution value.

The following is possible:

- Any number of FILESETs representable by a single character (A,B,C,D for instance)
- Not using a substitution character, rather extend the GWAFLIST so that each entry, rather than a single file name, represents the FILENAME, FILENAME FILESET(A), FILENAME FILESET(B), etc, etc
  - This would allow greater flexibility in the filenames used.
  - SMPFILE1-> £MPFILE1 or #MPFILE1 becomes SMPFILE1,ABCFILE1,DEFFILE1

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Some potential changes to meet alternative requirements

More than two FILESET solution

A slight variation of the two FILESET solution whereby two different underlying FILENAME definitions are defined, both of which are different to the FILENAME passed in by the application, is to have one of the underlying FILENAME definitions the same as the FILENAME passed in. This means that should the XFCREQ exit not be enabled, the application request will effectively default to the FILENAME specified in the application program.

This would add flexibility as to when the exit is enabled and allow the application to continue to run irrespective of the exit status.

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Some potential changes to meet alternative requirements

Timestamp switching

The default mechanism means that tasks check the FILESET to use when the task performs it first file request. This is because the exit isn't driven until a file request occurs for the task. If the GWA is updated to specify a new FILESET, all tasks starting after this event will use the updated FILESET value. Minor delays in updating this field could result in tasks in two different CICS regions being inconsistent.

The time-based switch option allows a precise switch over time to be specified in the GWA and the exit which could be running in multiple CICS regions, to precisely decide which FILESET to use based on the GWASWITCH time and the task start time.

The EXEC CICS INQUIRE ASSOCIATION STARTTIME() has been used to return the task start datetime in the format : YYYYMMDDHHMMSS.NNNNNN (20260209092610.658039 for example)

- The time part of this field has been used to determine the task's start time.
- The full field could be used if you wanted to switch based on date and time. An update to the GWA and sample programs SMPFCREQ and EXITMAIN would be needed.

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Further support

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IBM contacts

If you require further support with this sample. See the contact information provided in the GITHUB repository.

Additionally, you can contact ARNDT.EADE@UK.IBM.COM for general support and queries related to this solution.

Please note, this is a sample. So please thoroughly test your implementation.

Finally, feedback is always welcome. If you do implement a solution based on this sample, we'd love to hear from you.

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